2(a)(i) A man pushes a heavy rock resting on the ground, but it does not move. A student says that this is because the pushing force is balanced by the reaction to this force. Explain why the student's argument is wrong. [2]

(ii) Draw a labelled diagram to show the forces acting on the rock. [2]

(b) Block A of mass 2.0 kg moves with a velocity 10 m s^{-1} to the right on a smooth horizontal table. Block B of mass 3.0 kg moves with a velocity of 5.0 m s^{-1} in front of A in the same direction. A light spring of force constant 500 N m^{-1} is attached to B as shown in **Fig. 2.1** below.

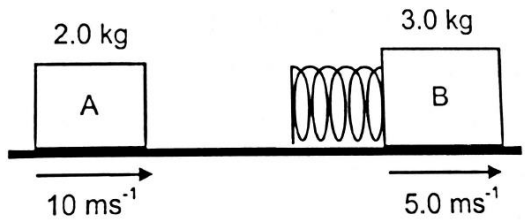

Fig. 2.1

(i) When A collides with B, there will be an instant when the spring experiences maximum compression. Describe the motion of A and B when maximum compression of the spring takes place. [1]

(ii) Calculate the final velocity of the two blocks at the instant the spring is at its maximum compression. [2]

(iii) Hence calculate the maximum compression of the spring. [2]